

JS

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JAVASCRIPT

COMPLETE BASIC NOTES

Learn • Practice • Create • Make Webpages Interactive

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1. What is JavaScript?

JavaScript (JS) is a programming language used to make webpages interactive and dynamic.

HTML creates structure, CSS designs the page, and JavaScript adds behaviour such as buttons, calculations, animations, forms and interactive games.



2. Why is JavaScript Used?

JavaScript can respond to user actions, change HTML and CSS, validate forms, perform calculations, create animations, build games and communicate with servers.

3. Advantages of JavaScript?

- Makes websites interactive.
- Runs directly in the browser.
- Can change webpage content.
- Useful for games and applications.
- Works with HTML and CSS.
- Can also be used for server-side development with Node.js.



4. How to Add JavaScript

Inline: JavaScript can be written inside an HTML attribute.

Internal: Write JavaScript inside the <script> tag.



External: Link a separate JavaScript file.

5. JavaScript Syntax

JavaScript statements are written as instructions for the browser.

6. Comments

Comments explain code and are ignored when the program runs.



7. Variables

Variables store data. Common keywords are **let**, **const** and **var**. Prefer **let** when a value changes and **const** when it should not be reassigned. `{ }`

8. Data Types

Basic types include String, Number, Boolean, Undefined, Null, Object and Array.

9. Operators

Arithmetic operators: +, -, *, /, %, **. Comparison operators include ==, ===, !=, !==, >, <, >= and <=. Logical operators are &&, || and !. `=>`

10. Input and Output

Use `alert()` for a popup, `prompt()` to receive text input, and `console.log()` to display information in the browser console. `{ }`

11. Conditional Statements

`if`, `else if` and `else` are used to make decisions. `</>`

12. Switch Statement

switch is useful when one value needs to be compared with several possible cases.



13. Loops

Loops repeat code. Common loops are for, while and do...while.

14. Functions

A function is a reusable block of code that performs a task.



15. Arrow Functions



Arrow functions provide a shorter function syntax.

16. Arrays

An array stores multiple values in one variable. Array indexes start from 0.

17. Array Methods

Useful methods include push(), pop(), shift(), unshift(), slice(), includes(), indexOf() and join().



18. Objects

Objects store information using key-value pairs.



19. Strings

Strings are text values. Common methods include `toUpperCase()`, `toLowerCase()`, `length`, `includes()` and `slice()`.

20. Numbers and Math

JavaScript provides Math methods such as `Math.round()`, `Math.floor()`, `Math.ceil()`, `Math.random()` and `Math.max()`.



21. Date and Time

The `Date` object is used to work with dates and times.



22. DOM – Document Object Model

The DOM allows JavaScript to access and change HTML elements.

23. Selecting HTML Elements

Common methods are `getElementById()`, `querySelector()` and `querySelectorAll()`.

24. Changing HTML and CSS

JavaScript can change text, HTML, styles and classes.

code



25. Events

Events happen when users interact with a webpage, such as click, mouseover, input, submit and keydown. `{ }`

26. Form Validation

JavaScript can check user input before a form is submitted.

27. setTimeout and setInterval

setTimeout runs code once after a delay. setInterval repeats code at a fixed interval. `=>`



28. Local Storage

localStorage can store small amounts of data in the browser.

29. Basic JavaScript Project – Digital Counter

This small project uses HTML, CSS and JavaScript to create a working counter.



30. Important Questions & Answers

Q1. What is JavaScript?

JavaScript is a programming language used to make webpages interactive.

Q2. What is the full form of JS?

JS means JavaScript.

Q3. What is a variable?

A variable stores a value that a program can use.

Q4. What is a function?

A reusable block of code that performs a task.

Q5. What is an array?

An array stores multiple values in one variable.

Q6. What is DOM?

DOM is a programming representation of an HTML document that JavaScript can manipulate.

Q7. Difference between let and const?

let can be reassigned; const cannot be reassigned.

Q8. What is an event?

An event is an action such as a click or key press that JavaScript can respond to.

31. Practice Questions

1. Write the full form of JS.
2. What is JavaScript used for?
3. Create a variable using let.
4. What are three JavaScript data types?
5. Write an if-else program.
6. Write a for loop from 1 to 10.
7. Create a function to add two numbers.
8. Create an array of five fruits.
9. What is DOM?
10. Create a button click event.

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JS

32. Complete Mini Project – Greeting Page

Practice HTML, CSS and JavaScript together.

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